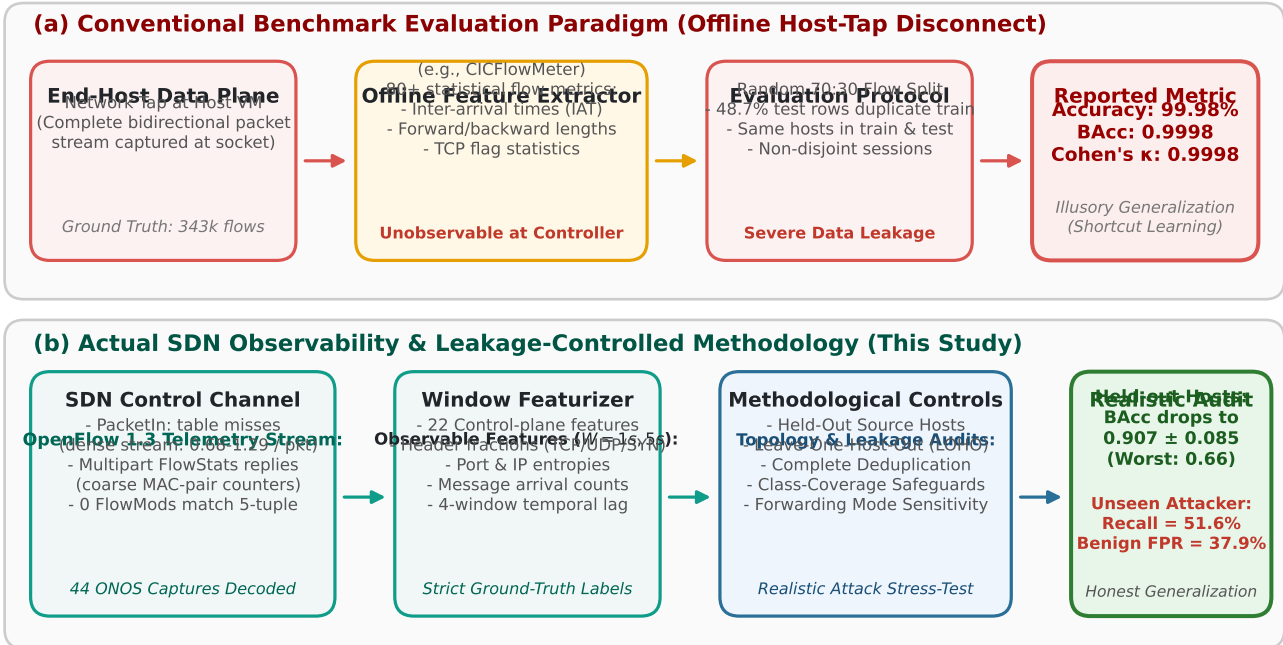

(a) Conventional Benchmark Evaluation Paradigm (Offline Host-Tap Disconnect)
End-Host Data Plane
(Complete bidirectional packet stream captured at socket)
Ground Truth: 343k flows
(e.g., CICFlowMeter)
80+ statistical flow metrics:
- Inter-arrival times (IAT)
- Forward/backward lengths
- TCP flag statistics
Unobservable at Controller
Random 70:30 Flow Split
- 48.7% test rows duplicate train
- Same hosts in train & test
- Non-disjoint sessions
Severe Data Leakage
Reported Metric
Accuracy: 99.98%
BAcc: 0.9998
Cohen's κ: 0.9998
Illusory Generalization
(Shortcut Learning)
(b) Actual SDN Observability & Leakage-Controlled Methodology (This Study)
SDN Control Channel
OpenFlow 1.3 Telemetry Stream:
- PacketIn: table misses
(dense stream: 0.66-1.29 / pkt)
- Multipart FlowStats replies
(coarse MAC-pair counters)
- 0 FlowMods match 5-tuple
44 ONOS Captures Decoded
Window Featurizer
Observable Features (W = 1s, 5s):
- 22 Control-plane features
Header fractions (TCP/UDP/SYN)
- Port & IP entropies
- Message arrival counts
- 4-window temporal lag
Strict Ground-Truth Labels
Methodological Controls
Topology & Leakage Audits:
- Held-Out Source Hosts
- Leave-One-Host-Out (LOHO)
- Complete Deduplication
- Class-Coverage Safeguards
- Forwarding Mode Sensitivity
Realistic Attack Stress-Test
Realistic Audit
BAcc drops to
0.907 ± 0.085
(Worst: 0.66)
Unseen Attacker:
Recall = 51.6%
Benign FPR = 37.9%
Honest Generalization